
Stablecoin Study Report

Stablecoins – Past, Present, and Future



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Stablecoins – Past, Present, and Future[☆]

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Abstract

Since 2018, stablecoins have become the major digital currencies facilitating payment flows across various decentralized financial platforms. While in the simplest terms, stablecoins operate as low-volatility digital cash, they are not created equal. Broadly speaking, stablecoins are dichotomized into fiat-backed or algorithmic; algorithmic refers to a diverse set of stablecoins. By reviewing a brief history and analyzing the current state of affairs for stablecoins, we discuss critical facts about stablecoins and propose strategic pathways for the future. Fiat-backed stablecoins are similar to money market funds in their function and structure. As such, the regulatory framework for money market funds should be a guiding light. From this viewpoint, the necessary conditions for the proper functioning of fiat-backed stablecoins are: transparency, full disclosure, quality auditing, and oversight. Algorithmic stablecoins are in principle synthetic cash, or a derivative instrument. As such, they are perfectly positioned to play pivotal roles in digital financial derivative (contingent claims) markets. Due to the heterogeneity of their use-cases, the appropriate regulatory framework depends heavily on the underpinning economics. Lending-related algorithmic frameworks (i.e., credit contingent claims) can benefit from economics and regulatory environments of collateralized repo markets and the like. Others can be considered through the lens of derivatives instruments—futures, options, swaps—for finding best practices and optimal strategies.

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Introduction

In January 2017, the market value of stablecoins (almost solely represented by Tether) stood at roughly US\$20M. By April 01, 2022, the total market cap of all stablecoins (more than 100) exceeded US\$180B.¹ Tether, the largest stablecoin, had a market cap of US\$83B. Terra, the third largest stablecoin, stood at a high of US\$18B. At the time of the writing of this report in June of 2022, Tether is down to US\$69B and Terra is defunct. The month of April 2022 will go down in history with infamy as the most tumultuous month for stablecoins when the world realized stablecoins may not be as stable as we believed. But is this as unexpected as we are made to believe? Were we misguiding ourselves that stablecoins are riskless? Lastly, are all stablecoins created equal? After all, some of the largest collateralized stablecoins, Circle's USDC and Binance USD, witnessed not only a premium in their conversion rates (vis-a-vis US\$), but also have seen dramatic gains in their market shares. Since April 2022, USDC has seen a more than US\$7B increase in its market cap, nearly equal to Tether's losses. We aim to shed light on some of the existing questions, and in so doing, we hope to pave the way for a more realistic understanding of stablecoins' forms and functions.

Where They Came From and Why

We start with: what are stablecoins? And why do we need them? Stablecoins are digital currencies that facilitate digital financial exchanges without the extreme volatility of prototypical cryptocurrencies. Cryptocurrencies were born out of the dire need for an effective means of digital exchange. Despite the ingenuity of their design, major cryptocurrencies like Bitcoin and Ethereum lack the price stability needed for widely accepted means of exchange. In its whitepaper, Centre, the consortium that governs USDC and was created by Circle and Coinbase, notes that "... existing public blockchain implementations and crypto assets struggle to fulfill the vision in part due to three significant challenges: price stability, transaction throughput, and risks due to the lack of independent governance over standards and network participants (particularly those members offering trade capability and fiat on- and off-ramps)."

In the most intuitive sense, then, stablecoins are digital money (albeit issued in a more decentralized way). While this definition lacks details, it does strike a chord in that for an average user, money should be seamlessly acceptable for any and all transactions. Stablecoins exemplify that ethos in digital finance.

¹ As of June 17, 2022, according to data from CoinMarketCap.com, the total market cap of the top 100 stablecoins were US\$157B with the top three (all of a collateralized variety) stablecoins—Tether, USDC and Binance USD—accounting for 90% of the total market capitalization. See: <https://coinmarketcap.com/view/stablecoin/>

This concept is for most part a shared understanding. According to CoolWallet, “[a] stablecoin is a cryptocurrency that remains stable in value against a pegged external asset class. It aims to minimize price volatility, by keeping its value fixed against that of a traditional real-world assets like a single fiat currency, a combination (basket) of currencies or a valuable real-life commodity. Its primary purpose is to negate the speculative nature of most cryptocurrencies and help create a more consistent and reliable market environment in order to increase adoption of digital assets.”^{2,3,4} In its “Digital Currencies: The Rise of Stablecoins”⁵ blog, the IMF notes that the rise of stablecoins is tantamount to “[a] battle [that] is raging for your wallet. New entrants want to occupy the space once used by paper bills or your debit card.”⁶

Current State of Stablecoins

While the IMF blog was written prior to the April 2022 crisis, some important points are worth noting. In its report, IMF notes six observations about stablecoins. First, stablecoin providers may steal away traditional deposits from banks. Second, tech giants can utilize their larger networks to create money monopolies. Third, stablecoins can replace weaker currencies as means of global exchange and payments. Fourth, stablecoins can make illicit trade easier. Fifth, stablecoins could provoke the loss of “seigniorage,” because central banks can capture profits from the difference between a currency’s face value and its manufacturing cost. And lastly, stablecoins can exacerbate financial consumer protection and financial inclusion concerns. For these reasons, the IMF calls for “[o]ne approach [that] would ... regulate stablecoins like money market funds that guarantee fixed nominal returns, requiring providers to maintain sufficient liquidity and capital.”

Given what has transpired since early April 2022, the above observations require careful consideration. Up until recently, banks have offered near-zero interest rates. Deposit insurance has never been sufficiently large for institutional clients. As such, stablecoins

² <https://www.coolwallet.io/the-complete-guide-to-stablecoins-part-i-2014-2019/>

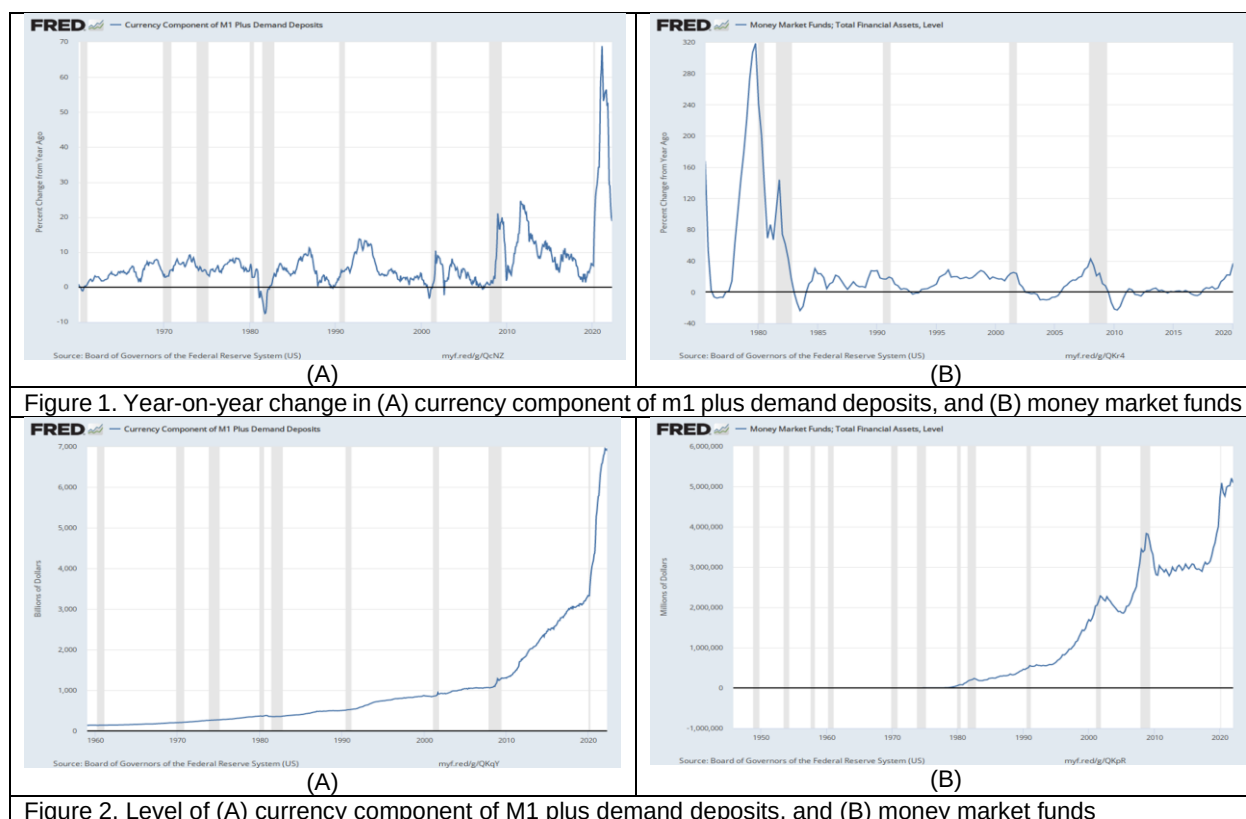
³ There is possibility for a self-reinforcing positive feedback loop. The innate stability, theoretically, can lead to diversifying and expanding the user base, thus boosting stablecoins’ utility for wider adoption. By creating more vibrant market dynamics, stablecoins in turn can strengthen stabilizing mechanisms such as cryptocurrency. Maybe, when cryptocurrencies become more stable, stablecoins will merge with more volatile cryptocurrencies, just becoming one of the varieties.

⁴ One can argue that price stability depends on the economic parameters of the stablecoin tokenomics model and market forces; transaction throughput is primarily determined by its technological implementation, while independent governance is a combination of a decision-making framework and processes enabling it to ensure that all stakeholders have power to influence decisions made.

⁵ <https://blogs.imf.org/2019/09/19/digital-currencies-the-rise-of-stablecoins/>

⁶ We clearly do not consider the case whereby different sorts of digital money (including CDBC’s and corporate private money) reside in wallets. What affects the convenience and utility of the end-user is beyond the scope of this analysis. We do not deny the possibility that user experience and competition among various digital monies have profound implications for each individual provider of such money.

do not necessarily pose a threat to prototypical bank deposits. Depending on the aggregate risk preference, stablecoins can be substitutes for money market funds or less risky alternatives to broader financial assets. As is evident in Figures 1 and 2, cash money increases during recessions whereas money market funds decrease. This suggests that if stablecoins (especially the fiat-backed variety) behave more like money market funds, there is no strong reason to believe they would hinder deposit taking functions of banks.



Furthermore, there is another type of cyclicity that influences the movement of money into and out of stablecoins. When the crypto market enters a downturn (a so-called crypto winter), theoretically the relative value of volatile cryptocurrencies should decrease vis-a-vis the value locked in stablecoins. The intuition is simple: if volatile cryptocurrencies lose value against fiat, then the value of any fiat-backed asset should remain intact and, in relative terms, stronger. As is shown in Figure 2, the relative value of stablecoins has actually increased against volatile cryptocurrencies. In 2017, stablecoins accounted for nearly 0% of the total crypto market cap. After two crypto winters (in 2018 and 2022), their market cap now stands at 17% of the total crypto market. This negative comovement is independent from any competition with bank deposits and is likely to remain a major driving force for the adoption of stablecoins.

Another important consideration is that stablecoins can actually help the traditional financial intermediaries grow. The dominant type of stablecoins by market cap is the fiat-backed stablecoins (e.g., Tether, USD Coin, and Binance USD) (See Figure 3). The fiat backing of these stablecoins ought to reside somewhere in the traditional financial system: bank deposits, money market funds, etc. Therefore, fiat-backed stablecoins do not compete with such traditional stores of money but rather create an intermediation layer between them and the stablecoin users.

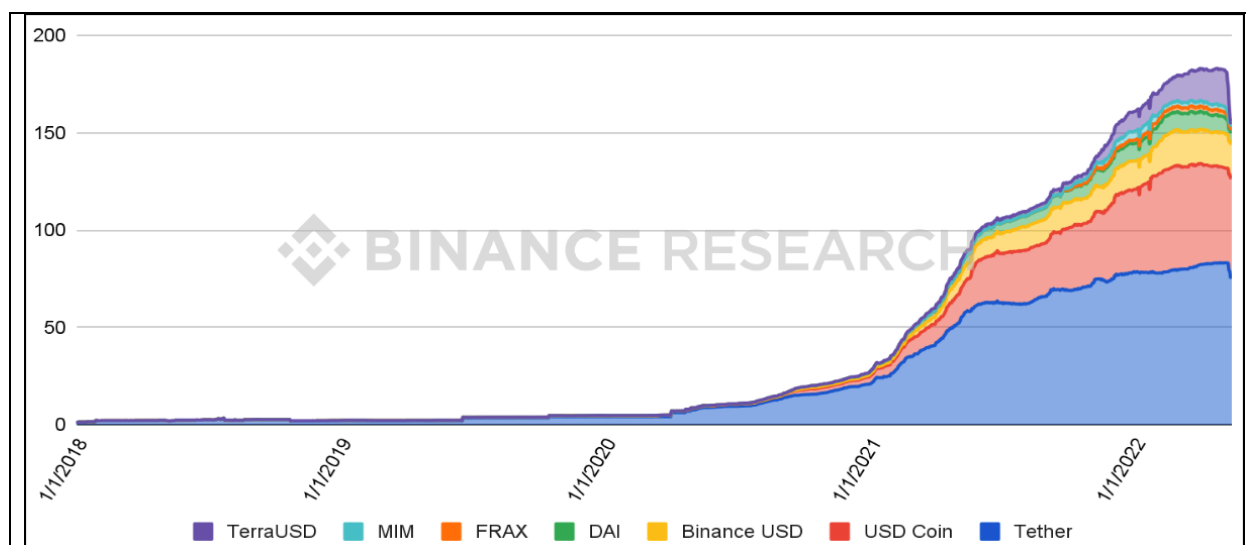


Figure 3. Market value of major stablecoins (2018-2022)

Tech giants such as Meta (formerly Facebook) have tried to make inroads into private currency. These attempts so far have been unsuccessful. Given the fact that the majority of cash resides in major financial institutions' balance sheets (Figure 4), perhaps it is not surprising that tech giants have not been able to succeed in creating private money. Since financial institutions remain the major source of systemic liquidity flow, tech giants find it difficult to replicate the legacy systems in their entirety.

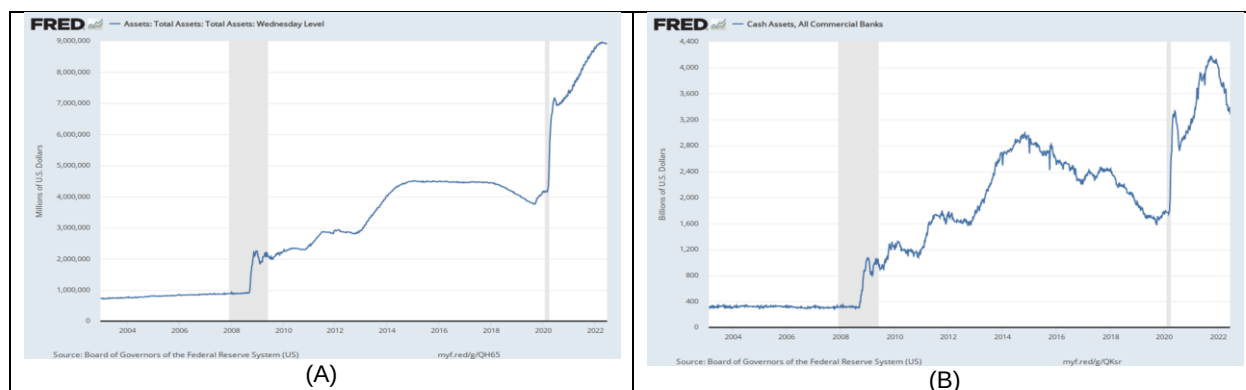


Figure 4. (A) All assets of Federal Reserve banks, and (B) Cash assets of all commercial banks

In the case of cross-border payments, stablecoins may have a theoretical advantage. However, any cross-border payment faces the classic “impossible trilemma.”⁷ As economists John Marcus Fleming and Robert Alexander Mundell demonstrate, in international open economies, it is impossible to have all three of these facets at the same time: (1) a fixed foreign exchange rate, (2) free capital movement (absence of capital controls), and (3) an independent monetary policy. Depending on what parts of the trilemma a country intends to control, the cross-country stablecoin solution may or may not prove effective. Nonetheless, there is no clear answer as to whether cross-border payment will be assisted or hindered by stablecoins.

Stablecoins pegged to stronger national currencies do indeed present a challenge for central banks of weaker national currencies. For instance, stablecoins allow people in Argentina and Russia to bypass rules that limit or tax the holding of USD and other foreign currencies in their bank accounts. This probably explains why USD-pegged stablecoins can be sold in peer-to-peer exchanges for a price greater than the price of USD in RUB.

The idea that stablecoins may ease illicit trades stems from the fact that transactions on a blockchain are pseudonymous. Compared to transactions via commercial banks and other financial institutions, where the identities of the parties involved in the transactions are known by the financial institutions and these financial institutions are obliged to conduct KYC/AML/CFT diligence on these transactions, it is clear that blockchains, and by extension stablecoins, can increase risks of illicit transactions. However, it is also worth noting that: a) transactions on a blockchain, despite their pseudonymity, are completely public, providing a degree of transparency that is greater than in non-blockchain-based payment infrastructures; b) this greater transparency can be and is being used to monitor transactions and fight financial crimes, as exemplified by Chainalysis; c) prototypical users will eventually have to convert stablecoins to fiat currency (i.e., the so-called “crypto off-ramp” process) to use the money within the non-crypto economy, and this process occurs through regulated financial institutions that must comply with KYC/AML/CFT; d) fiat-backed centrally-operated stablecoins typically use smart contracts that give the operating entity the power to freeze the balances of users, and this power has already been used to fulfill requests from financial authorities; e) the level of anonymity and privacy achievable with transactions on a blockchain is actually lower than that of physical cash. Transactions can be traced to the point of origin addresses, especially when corresponding hard wallets are in use.

Whether stablecoins provoke the loss of “seigniorage” is an empirical question yet to be answered, and it is likely that the answer will depend largely on predominant types of stablecoins. Fully-backed, fiat-backed stablecoins, for instance, are theoretically

⁷ https://en.wikipedia.org/wiki/Impossible_trinity

incapable of provoking seigniorage losses, because any amount of stablecoin issued needs to be backed by an equivalent amount of fiat currency, and seigniorage will indirectly arise from issuing the fiat currency necessary for backing. In fact, if a fiat-backed stablecoin is fully backed by the currencies issued by central banks, it may even lead to a gain of seigniorage compared to commercial bank money, which does not need to be fully backed by central-bank money within the fractional reserve banking system. On the other hand, stablecoins that are unbacked, only fractionally backed, or backed by assets that are not issued by a central bank can indeed reduce a central bank's seigniorage. Particularly in the case of crypto-backed stablecoins, it could be argued that seigniorage is being shifted away from central banks and to the entities that benefit from the increasing supply of the backing cryptocurrency. Such entities are, for instance, "miners" in the case of proof-of-work cryptocurrencies and block producers and validators in the case of proof-of-stake cryptocurrencies.

The global concerted effort to launch central bank digital currencies, at least at face value, suggests that major monetary authorities prefer to have their own digital money. The concerns about consumer access to financing services and privacy are a mixed bag. So far the largest hacks and thefts occurred on exchanges (e.g., KuCoin in 2020, Upbit in 2019, BINANCE in 2019). The flaws in security designs of the exchanges seem to be the main culprit. We really cannot blame stablecoins for that. Similarly, the consumer access question is also intimately related to decentralized applications (dApps). The diversity and design of dApps determine whether the ecosystem is consumer friendly. The variety of financial services is germane for the success of the ecosystem but stablecoins only facilitate movement of money. In short, stablecoins grease the wheels, but which direction the wheels turn is another question altogether.⁸

Lastly, stablecoins are also a solution for an interoperability problem. Blockchains offer programmability. There are now numerous decentralized finance applications running almost autonomously on blockchains, offering a wide range of financial products and instruments such as staking, lending and borrowing, options, futures, prediction markets, crowdfunding and launchpad platforms. On these applications, end-users can use their digital assets but they cannot use their fiat currencies directly, simply because fiat currencies do not exist on blockchains. Stablecoins bridge this interoperability gap by giving people assets that are pegged to fiat currencies and that can be used within this

⁸ An interesting related question, which we intentionally sidestep for future discussion, is: whether stablecoin users can also shape the directions of various ecosystems? Market demand is clearly an indicator of consumer preferences. Digital assets in general and stablecoins in particular have other technological capabilities, especially through smart contracts, that are embedded in their design. Can consumers voice their opinion not only via their spending power but also via voting or metadata sharing mechanisms that can be connected to a token or a wallet? We do see these issues as critical, deserving dedicated analyses.

new world of blockchain-based finance. From this point, stablecoins do not compete with fiat currencies or even CBDCs, but rather complement them by making them available within blockchain ecosystems. Indeed, this point stems from our earlier, simple, yet fundamental assertion that stablecoins are digital cash for digital economies.

Stablecoin Classification

In light of recent collapses and regulatory movements, it is crucial to realize that there are many different types of stablecoins and they should not all be treated in the same way. Indeed, Thomas Cowen from Paxos notes that "... Not all stablecoins are equal or even 'stable.' ... Fiat-backed stablecoins are backed by just that—fiat-denominated traditional currencies and assets. However, different fiat-collateralized stablecoins are backed by different assets depending on risk appetite."⁹ One important dimension of classification is whether the stablecoin is "algorithmic." Although this word is often used to refer to stablecoins that are unbacked and uncollateralized, it actually just means that stablecoin's stabilization mechanisms follow well-defined rules, usually programmed in a smart contract. There are plenty of backed or collateralized algorithmic stablecoins, of which Djed and DAI are two examples.

Backing and collateralization are often conflated and their subtle difference is a distinguishing classification factor that is frequently overlooked. When a stablecoin is backed, it has a reserve and can be redeemed for portions of the reserve. When it is collateralized, it can be minted in the form of loans against provided collateral, but the collateral can only be retrieved by the user who generated the loan when he or she pays it back or in the event of liquidation.

Another important criterion for classification is governance. At one end of the spectrum, we have fiat-backed non-algorithmic stablecoins, which are governed by a single company. At the other end of the spectrum, we have fully autonomous algorithmic stablecoins, where the smart contracts run completely independently and immutably and are not governed by anyone. Somewhere in the middle of the spectrum, we may have algorithmic stablecoins that have an associated governance token that gives rights to a wide group of people to participate and influence decisions that may change aspects of the algorithm. Table 1 provides a broad classification of major stablecoins.

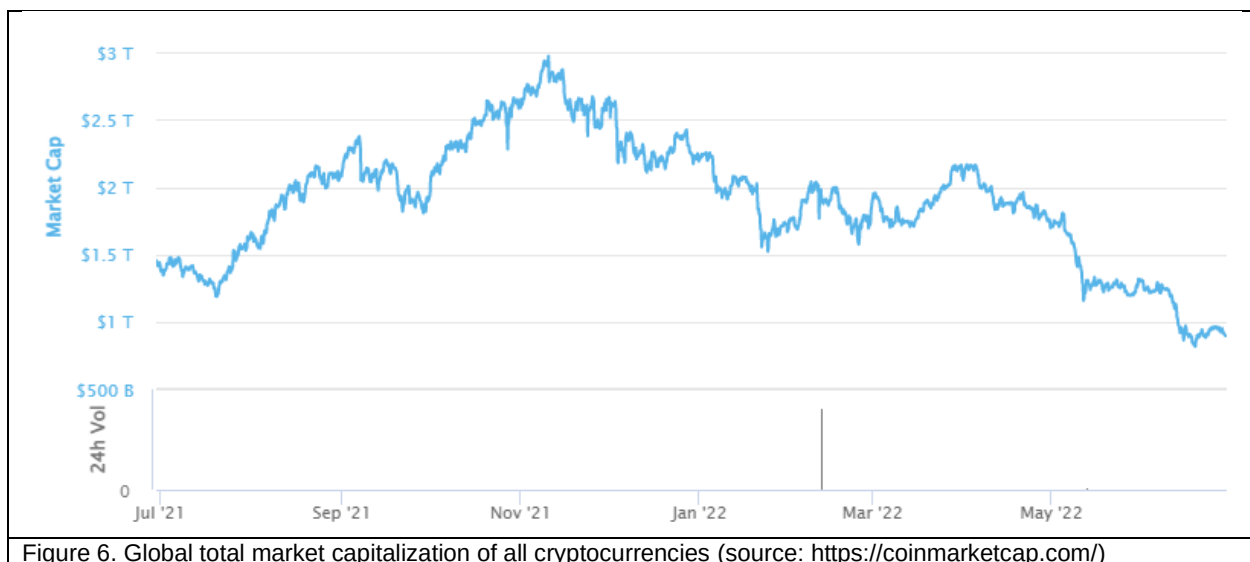
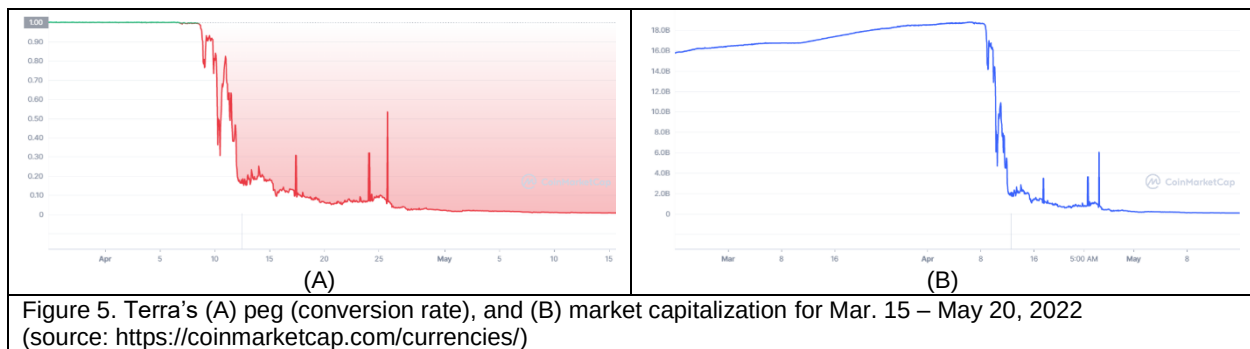
Table 1. Top 10 Stablecoins by Market Capitalization (source: https://coinmarketcap.com/view/stablecoin/)				
Logo	Name	Symbol	Market Capitalization (US\$)	Type
	Tether	USDT	68,078,303,248	Backed by USD
	USD Coin	USDC	55,386,709,457	Backed by USD

⁹ <https://paxos.com/2022/07/13/its-not-either-stablecoins-or-cbdcs-its-both/>

	Binance USD	BUSD	17,426,551,123	Backed by USD
	DAI	DAI	6,743,975,639	Algorithmic; over-collateralized by ETH, USDC and other assets
	Frax	FRAX	1,422,834,141	Algorithmic; partially backed by USDC
	True USD	TUSD	1,230,789,899	Backed by USD
	Pax Dollar	USDP	943,379,977	Backed by USD
	Neutrino USD	USDN	787,727,754	Algorithmic; backed by WAVES
	Liquity USD	LUSD	730,074,552	Algorithmic; collateralized
	USDD	USDD	693,596,276	Semi-algorithmic

The Collapse of Terra

In April/May 2022, the Terra stablecoin collapsed (Figure 5). The cascading effect was an almost 50% reduction in the total market cap of the cryptocurrency industry (Figure 6) and the bankruptcy of many companies. The pertinent question is: what happened?



To answer this question, we examine whether the claim that Terra was an algorithmic stablecoin was valid. As a stablecoin, Terra's algorithmic design should have brought about economic conditions and incentives to make it secure. Terra's failure then points to flaws in the algorithm. But what were these flaws?

Terra's algorithm was originally just a variation of the seigniorage shares stablecoin algorithm,¹⁰ where the LUNA token played the role of the seigniorage share. The intuition is simple. When the value of UST is above 1 USD, LUNA holders can burn 1 USD worth of LUNA tokens in exchange for 1 UST. Then they can sell this 1 UST in the market for a value between the current market price and 1 USD, thereby profiting (in the form of indirect seigniorage revenue) and contributing to an increase in the supply of UST and, consequently, a reduction of its price in the market. Conversely, when the value of UST is below 1 USD, UST holders can burn UST in exchange for 1 USD worth of LUNA. This decreases the supply of LUNA in the open market and increases its price.

This idea seems convincing at first sight and because one can exchange UST for LUNA, UST appears to be reassuringly "backed" by LUNA. However, it is incorrect to think of this possibility of exchange as a proper form of backing; the value of LUNA is dependent on the expected future demand for the stablecoin. It is a circular dependency, which entails that UST is "backed" by itself and hence not really backed by anything else. When UST collapsed, the price of LUNA fell as well due to this circular dependency. This problem is not exclusive to Terra. It is inherent to pure seigniorage share-style algorithmic stablecoins: when the stablecoin loses the peg significantly, the seigniorage share that was allegedly "backing" it will also lose value since there won't be sufficient demand for a stablecoin that is not stable and hence the expectation of seigniorage inflow will be low.

Proper backing necessitates using assets that are independent from the stablecoin protocol itself. Thus it is more appropriate to refer to pure seigniorage share-style stablecoins as unbacked. It seems that the Terra project even realized this towards its end since a backing reserve in Bitcoin was created by the Luna Foundation Guard to be non-algorithmically managed and governed. However, this reserve did not sufficiently (fully) back all UST in circulation. Furthermore, it is also possible that the purchase of Bitcoins may have been just an exit before and during the collapse by insiders.

Another problem of seigniorage share-style stablecoins is that seigniorage, which is revenue originating from increasing the supply, creates a perverse incentive to increase the supply. To increase the supply, demand for the stablecoin needs to exist. In the case of Terra, this demand was artificially generated by an annual percent rate of almost 20%

¹⁰ <https://blog.bitmex.com/wp-content/uploads/2018/06/A-Note-on-Cryptocurrency-Stabilisation-Seigniorage-Shares.pdf>

for locking UST on Anchor, a lending and borrowing platform on top of Terra. At one point in time, almost 80% of the entire UST supply was deposited in Anchor. If for any reason, including but not limited to unsustainability of 20% rate, these vast amounts of UST were withdrawn from Anchor and sold in the market, there would clearly be a sudden increase in the circulating supply and a corresponding sudden crash in the UST price. A sensible course of action would have been to reduce Anchor's rate to encourage its latent supply to be withdrawn gradually. But Terra did exactly the opposite. Several weeks before the collapse, it bailed out Anchor. This decision, which goes contrary to stability, is understandable from the point of view of the perverse seigniorage incentive. The steep increase in the total supply of UST that correlated with the steep increase of deposits in Anchor had certainly been lucrative for LUNA holders so far.

Any stablecoin can be subject to a sudden supply increase and corresponding price crash, but good algorithmic stablecoins would be able to absorb the shock by buying back the excess supply. In the case of Terra, due to the joint crash of LUNA for the reasons explained above, buying back the excess supply was not possible. Good algorithmic stablecoins would also have incentives that promote stability instead of perverse incentives that encourage behaviors that cause instability.

Another issue with Terra was related to governance. When UST and LUNA crashed, Terra halted the blockchain, thereby manually preventing the algorithm from working. This is the opposite of what an "algorithmic" stablecoin should do. They were able to do this because they controlled the nodes of the blockchain. Their blockchain was not independently operated. If they had allowed the algorithm to work, UST holders would be able to exchange their UST for vast amounts of LUNA tokens, which would be dilutive and hence not beneficial to the incumbent LUNA holders. So naturally, it was not in their interest to allow the algorithm to work, even if doing so would have provided some support for the price of UST.

Future Directions

Challenges of stablecoins as private digital money are not trivial. Gorton, Ross, and Ross (2022) further show that "It is difficult for private agents to produce money that circulates at par with no questions asked. We study two cases of privately-produced money: pre-Civil War U.S. private banknotes and modern stablecoins. Private monies are introduced when there are no better alternatives, but they initially carry an inconvenience yield. Over time, these monies may become more money-like, but they do not always achieve a positive convenience yield. Technology advances and reputation formation pushed private banknotes toward a positive convenience yield. We show that the same forces

are at work for Stablecoins.”¹¹ This implies that while there are no better alternatives to function as digital money, stablecoins can enjoy a collective monopolistic rent. However, the growing number of central bank digital currencies (CBDCs) is a major potential competitive threat to private money. Of course as Gorton, Ross, and Ross (2022) note, both CBDCs and stablecoins can play complementary roles. After all, money market funds and bank deposits co-exist and provide complementary services. However, building on the money market fund analogy, the emerging CBDCs may pave the way for collateralized stablecoins to function as digital private money.

More importantly, an observation of the peg and market capitalization of top stablecoins reveals stark patterns [Figures 7A to 7G]. Since April 2022, only USD Coin (USDC) has gained market share in a dramatic way. The largest, Tether, has seen a drastic decrease in its market share. According to a *New York Times* article,¹² “... [a]s cryptocurrencies plummeted, a flood of investors asked to exchange their Tethers for dollars, forcing the company to pay out about an eighth of its reserves, or \$10 billion, over the course of a week and a half. On cryptocurrency exchanges, Tether briefly wavered from its \$1 peg.” Two notable algorithmic coins, DAI and FRAX, have also lost significant market share since April 2022. By all accounts, however, backed and collateralized coins have fared well; furthermore, among these, coins with reserve or collateral assets that were perceived as less risky have fared even better (especially USDC).



Figure 7A. Tether's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022



Figure 7B. USD Coin's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022

¹¹ “Making Money,” January 2022, Gary B. Gorton, Chase P. Ross & Sharon Y. Ross, NBER working paper 29710; DOI 10.3386/w29710; see: <https://www.nber.org/papers/w29710>

¹²Tether: The Coin That Could Wreck Crypto - *The New York Times*

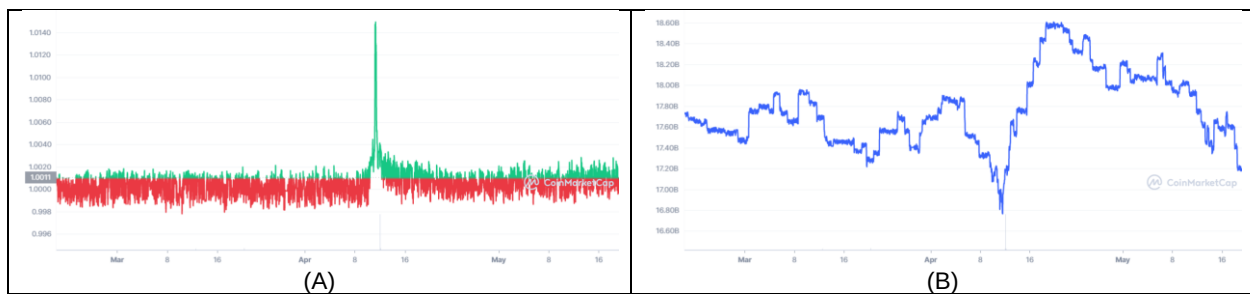


Figure 7C. Bianco USD's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022

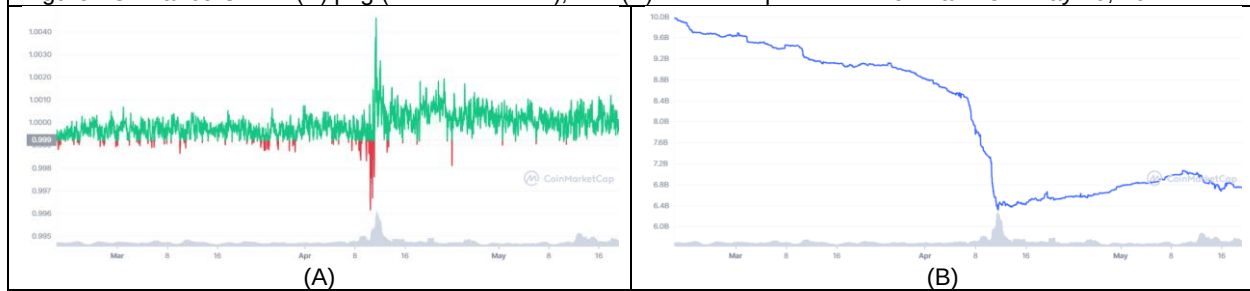


Figure 7D. DIA's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022



Figure 7E. FRAX's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022



Figure 7F. True USD's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022



Figure 7G. Pax Dollar's (A) peg (conversion rate), and (B) market capitalization for Mar. 15 – May 20, 2022
(source: <https://coinmarketcap.com/currencies/>)

In short, we can see two potential future pathways:

- Fiat-backed centrally operated stablecoins can fill a void in the digital financial services much the same way as money market funds have. But to do so, these stablecoins can take a page from money market funds' history and (1) define explicitly without ambiguity their investment strategy, (2) disclose fully the portfolio of assets that back the stablecoins, (3) adhere to highest quality auditing standards, and (4) welcome regulatory frameworks that provide clarity and protections for the well-functioning of their respective industries. Given the critical role money market funds have played in revolutionizing institutional investment, corporate treasury management, and wealth advisory services, we tend to think regulatory separation of fiat-backed stablecoins can usher in an age of near-safe, fully digital private money that can harmoniously coexist with future CBDCs. This private digital money will facilitate digital exchange needed by a fully functioning digital economy.
- Algorithmic stablecoins can continue to explore the programmability of money made possible by blockchains and smart contracts in order to automate stabilization mechanisms and ultimately offer stable digital assets with less operational overhead. To do so while having the confidence of users, these stablecoins must (1) define explicitly the stabilization mechanisms, (2) argue with more than just informal explanations why these mechanisms work, (3) be clear about the risks involved, and (4) optimize governance to limit perverse incentives of governance token holders. Given the importance of artificial intelligence, automation, and autonomy in multiple industries, it seems plausible and indeed probable that autonomy in the minting and burning of programmable digital money through smart contracts will be an inevitable part of our future. A favorable regulatory environment that enables innovation is essential for the full realization of potential algorithmic stablecoins.

The dichotomy provides a much needed pathway for algorithmic stablecoins to play a vital yet less explored role within the digital finance ecosystem. In the parlance of finance, these instruments are derivatives: they are meant to synthetically mimic the movement of their underlying assets (fiat, crypto, etc.). Since the first modern derivatives (options) started trading in the early 1970s, financial derivatives have become the largest segment of asset classes. But more importantly, they have provided innovative means for financial engineering. A case in point is the story of portfolio insurances. After the Nobel prize winning works of Black-Scholes-Merton on option pricing and proliferation of option trading in the 1970s, the decade of the 1980s brought about a new breed of derivatives: synthetics. By the early 1980s it became very clear that options have tremendous potential for risk mitigation. Put options—the bet on a price decline—were especially useful in insuring against downturns. However, outright put options are expensive insurance. American financial economists Hayne Leland and Mark Rubinstein thus

invented synthetic puts. Using other cheaply available derivatives and assets, they replicated a dynamically adjusting portfolio insurance. What followed is nothing short of a miracle. The algorithmic trading accounts for the majority of the order flow and credit default swaps that dwarf the bond market are only two prominent examples of how synthetic financial products have changed the landscape of finance.

By dichotomizing stablecoins along their ultimate functions, which happens to coincide with their form, we can unleash the true potential of these products. Moreover, this change of perspective can also lead to better alignment with regulatory frameworks and disclosure rules. From a practical standpoint, this can also result in more effective risk management. Take for instance counterparty risk, which has been perhaps the singular critical reason for the recent failings in stablecoins. Fiat-backed stablecoins, which are backed by high quality, high visibility assets, should not in theory face dire credit/counterparty risk. However, as the global financial crisis of 2008 demonstrated, even the safest instruments—money markets—can face occasional bouts of extreme illiquidity shocks. The fiat-backed stablecoins thus need to have in place effective mechanisms to mitigate liquidity risk. Equally importantly, for the purpose of adhering to AML-KYC and other regulatory requirements, these stablecoins need to build strict operational risk management systems that may sacrifice flexibility expected from decentralized systems to provide soundness.

For algorithmic stablecoins, the inherent risks associated with synthetic derivatives demand sophisticated financial engineering and multi-faceted risk management mechanisms (credit, counterparty, liquidity, operational, and regulatory). But equally importantly, the success and growth of these stablecoins necessitate the lock-step development of a broad financial system. In its fully developed state, this financial system includes essential elements such as account insurance, clearinghouses, risk-based prudential capital (equity) requirements, and intersystem connectivities. Given that algorithmic stablecoins rely on multilateral financial assets' trading to maintain stability, they must implement optimal trading mechanisms that not only facilitate speedy transactions but also provide necessary backstops to prevent panic trading and suboptimal liquidations. The interconnectivity to lending platforms and leveraged instruments exacerbates the fragility of financial systems predicated on algorithmic trading. From both business longevity and community welfare standpoints, algorithmic stablecoins are well-served in implementing and maintaining a wholesome approach to risk management that expands beyond immediate peg stability and transcends to the entirety of the financial ecosystem they serve.

Conclusion

Stablecoins are necessary means of exchange in digital economies. Despite the failings of a few operators in this nascent industry, digital financial systems offer extraordinary opportunities: speedy transactions, access for underbanked and unbanked, and democratized exchange and trading ecosystems. However, to reach their full potential, we must first recognize that the form of stabilization mechanism fundamentally changes how stablecoins should be managed from a business perspective and be regulated from an oversight standpoint. We propose that fiat-backed stablecoins are best viewed through the lens of money market funds within traditional financial services. These stablecoins can play a similarly pivotal role in the development of digital financial services. To do so, full backing by risk-free assets, complete transparency, and audited reporting are must-haves. On the other hand, the algorithmic stablecoins are uniquely positioned to serve as synthetic cash in expanded financial ecosystems that include a wide swath of financial instruments. Prudent capital requirement, effective wholesome risk management, and liquidity shock-proof trading mechanism are critical necessary conditions for creating fragility-resistant financial systems based on algorithmic stablecoins. We hope that this document will pave the way for more constructive dialogue on the form, function, and future of stablecoins.

Appendix A – Acknowledgements

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Bruno Woltzenlogel Paleo is the architect of the Djed algorithmic stablecoin protocol and the founder of the Djed Alliance. Previously he was Director of Engineering and Technical Director for Stablecoins and CBDCs at Input Output, the company that builds the Cardano blockchain. He holds a Ph.D. in Computer Science from the Vienna University of Technology.		Rodney Prescott has been involved with blockchain for over 10 years. He has worked for EY and PWC in blockchain, as well as for the digital asset custody firm, QREDO. Recently Rodney has been CTO for a regulated Crypto Brokerage. He is currently undertaking a Doctorate In Business Administration, focusing on the financial management of digital assets.	